

Phase Four: Collect the Data.

● FIRST : Clean Data Set.

We clean data set from arabic and Non-Carelessness responds .

● SECOND : Fill the Missing Values

1. Here we load our dataset of responses of the survey

```
import numpy as np
import pandas as pd
df = pd.read_excel('C:\\Users\\User\\Desktop\\responses.xlsx')
print(df)
```

```
What's your Name \
0  2023-04-27 20:54:42.391  0  Salma Soliman
1  2023-05-01 18:07:54.665  0  Aya Ahmed Rehan
2  2023-05-01 18:19:54.363  0  Mariam
3  2023-05-01 19:13:09.569  0  Tasneem Rehan
4  2023-05-01 20:51:19.904  0  Shahd asaya
..  ...
521 2023-05-07 20:54:58.056  0  Ahmed desoky
522 2023-05-07 21:02:05.498  0  The big M
523 2023-05-07 21:14:07.762  0  NOURHAN
524 2023-05-07 21:38:44.041  0  Mona Badr
525 2023-05-07 22:35:10.737  0  Yasser Fouad
```

```
Which of the following is your favourite cuisine \
0  Egyptian food
1  Egyptian food
2  Italian food, Egyptian food
3  Egyptian food
4  Egyptian food
..  ...
521  Egyptian food
522  Italian food, Egyptian food, Lebanese food
523  Italian food
524  Egyptian food
525  Chinese food
...
524  0 L.E
525  NaN
```

[526 rows x 32 columns]

2. Check the number of NAN values in each column

```
df.isnull().sum()
```

13]

```
..  0
    0
    ?What's your Name 6
    Which of the following is your favourite cuisine 0
    ?What's your favourite chef 132
    ?How do you define your food regime 0
    ?How often do you go shopping for food 34
    ?How would you describe your diet 0
    Is your country famous for any foods ? What are they 73
    What meal do you prefer eating through day ? 0
    Are you allergic to any of the following food items?(please select all that apply) 137
    ?Which do you prefer for breakfast 0
    Choose What kind of food you consume regularly for lunch 0
    How much do you prefer salt in your food? 3
    ?Do you like spicy food 0
    ?What do you prefer for dinner 2
    Do you eat various types of vegetables or fruit? 7
    If you eat vegetables or fruits mention what you prefer? 3
    Please mention which ingredients that you dislike in meals 112
    ? How often do you consume fast food 5
    What kind of fast food do you usually consume? 0
    Please select your favourite restaurant 3
    ?How do you like your food 0
    Describe reasons why people enjoy fast food 7
    Which dessert of the following you prefer? 0
    ...
    ?What is your gender 0
    ?What's your age 0
    ?What's your education 0
    ?How much is your monthly income 34
    dtype: int64
```

تابع زمني
النتيجة

3. Replacing NAN values with the most frequent occurred item in each column using `fillna()` function which iterates through our dataset and fills all empty rows with a specified value.

```
df = pd.read_excel('C:\\Users\\User\\Desktop\\responses.xlsx')
# numeric columns
for col in df.select_dtypes(include=['number']):
    df[col].fillna(df[col].mean(), inplace=True)

# categorical columns
for col in df.select_dtypes(include=['object']):
    df[col].fillna(df[col].mode()[0], inplace=True)

print(df)
```

4]

	النتيجة	طابع زمني	What's your Name \
0	2023-04-27 20:54:42.391	0	Salma Soliman
1	2023-05-01 18:07:54.665	0	Aya Ahmed Rehan
2	2023-05-01 18:19:54.363	0	Mariam
3	2023-05-01 19:13:09.569	0	Tasneem Rehan
4	2023-05-01 20:51:19.904	0	Shahd asaya
..	...	...	...
521	2023-05-07 20:54:58.056	0	Ahmed desoky
522	2023-05-07 21:02:05.498	0	The big M
523	2023-05-07 21:14:07.762	0	NOURHAN
524	2023-05-07 21:38:44.041	0	Mona Badr
525	2023-05-07 22:35:10.737	0	Yasser Fouad

	Which of the following is your favourite cuisine \
0	Egyptian food
1	Egyptian food
2	Italian food, Egyptian food
3	Egyptian food
4	Egyptian food
..	...
521	Egyptian food
522	Italian food, Egyptian food, Lebanese food
523	Italian food
524	Egyptian food
525	Chinese food

4. Finally we check again the NaN values

```
df.isnull().sum()
```

```
0
0
?What's your Name 0
Which of the following is your favourite cuisine 0
?What's your favourite chef 0
?How do you define your food regime 0
?How often do you go shopping for food 0
?How would you describe your diet 0
Is your country famous for any foods ? What are they 0
What meal do you prefer eating through day ? 0
Are you allergic to any of the following food items?(please select all that apply) 0
?Which do you prefer for breakfast 0
Choose What kind of food you consume regularly for lunch 0
How much do you prefer salt in your food? 0
?Do you like spicy food 0
?What do you prefer for dinner 0
Do you eat various types of vegetables or fruit? 0
If you eat vegetables or fruits mention what you prefer? 0
Please mention which ingredients that you dislike in meals 0
? How often do you consume fast food 0
What kind of fast food do you usually consume? 0
Please select your favourite restaurant 0
?How do you like your food 0
Describe reasons why people enjoy fast food 0
Which dessert of the following you prefer? 0
...
?What is your gender 0
?What's your age 0
?What's your education 0
?How much is your monthly income 0
dtype: int64
```

Collected data phase (Age & Education cleaning) :

First of all we deleted all string values we didn't need from column "Age"

```
df["Age"]=df["Age"].str.replace("[YearsYearyearsoldN?MatiamyaerGoodPeter]", "",
regex=False)
```

After that we converted age column to be numeric value

and then we filled any null values using the mean age

```
df['Age'] = pd.to_numeric(df['Age'], errors='coerce')
df['Age'].fillna(df['Age'].mean() , inplace=True)
```

We filled any age under 18 using mean imputation

```
for i in range(len(df)):
    if df.loc[i,"Age"] < 18:
        df.loc[i,"Age"] = df['Age'].mean()
```

After that we converted age values to be integers

```
df['Age'] = df['Age'].astype(int)
```

We made a loop to fill and correct the level education for the participant based on his/her age

```
for i in range(len(df)):

    if df.loc[i,"Education"] == "Teaching assistant" and
(df.loc[i,"Age"] <23 or df.loc[i,"Age"] > 30):
        df.loc[i,"Education"] =None

    elif df.loc[i,"Education"] == "Doctor" and df.loc[i,"Age"] < 25:
        df.loc[i,"Education"] =None
```

```
elif df.loc[i,"Education"] == "Level 2" and df.loc[i,"Age"] < 19:
    df.loc[i,"Education"] = None

elif df.loc[i,"Education"] == "Level 3" and df.loc[i,"Age"] < 20:
    df.loc[i,"Education"] = None

elif df.loc[i,"Education"] == "Level 4" and df.loc[i,"Age"] < 21:
    df.loc[i,"Education"] = None

elif df.loc[i,"Education"] == "Level 5" and df.loc[i,"Age"] < 22:
    df.loc[i,"Education"] = None

elif df.loc[i,"Education"] == "Level 6" and df.loc[i,"Age"] < 23:
    df.loc[i,"Education"] =None
else:
    df.loc[i,"Education"] = df.loc[i,"Education"]
```

Finally, we cleared any null values from "Education" column.

```
df.dropna(subset=["Education"], inplace=True)
```